Modeling Disorder in a Binary Fe–20 at.% Al Alloy

Cynthia Wen
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ETH Zürich, Department of Materials

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Abstract

Atomic disorder controls many technologically important properties, from the thermoelectric efficiency of half-Heuslers to the relaxor behavior of ferroelectrics. Such disorder can be measured with single-crystal diffuse scattering from X-ray and neutron diffraction, but interpreting these signals remains difficult, as contributions from short-range order (SRO), size-effect strain, and thermal diffuse scattering overlap in a single diffraction map. Here, a workflow is proposed to separate these contributions using a binary Fe–20 at.% Al alloy as a model system. Chemical SRO correlations are refined against synchrotron data using the three-dimensional difference pair distribution function (3D-ΔPDF). Shell-resolved interaction parameters are then fitted to reproduce them by histogram reweighting and used to drive Monte Carlo simulations of a 16,000-atom crystal. The static displacements that give rise to the size effect are introduced by relaxing these configurations with a universal machine-learned interatomic potential. Pre-trained on large quantum-mechanical databases, such potentials reach near density-functional accuracy at a cost low enough to relax tens of thousands of atoms, removing the scaling barrier that has limited first-principles relaxation in disordered crystals. Thermal motion is included only through an isotropic approximation, leaving explicit modeling of correlated vibrations to future work. The resulting model reproduces the experimental diffuse scattering well across multiple reciprocal-space planes, and this approach should extend to more complex materials with similar disorder.

1 Introduction

Material properties are often controlled not only by the average structure, but also by deviations from it, i.e., by disorder [1]. This disorder can be captured experimentally with X-ray diffraction or neutron scattering [2, 3], and appears as diffuse scattering between and beneath Bragg peaks. Extracting structural information from these signals is challenging, however, as measured diffuse scattering contains overlapping contributions from chemical short-range order (SRO), static size-effect displacements, and thermal vibrations [4].

Understanding these signals is essential because local structure governs many technologically important properties [5]. For example, in relaxor ferroelectrics such as $Pb(Mg_{1/3}Nb_{2/3})O_3$ (PMN), chemical SRO plays a role in determining the size of polar nanoregions [6], which can enhance piezoelectric effects [7], important for ultrasound and sonar applications [8]. In half-Heusler materials such as $Nb_{1-x}CoSb$, local vacancy order determines the thermoelectric figure of merit [9], which sets how efficiently the material converts waste heat into electrical energy [10].

Both these examples contain a binary alloy substructure, and alloy-based modeling methods have been applied to both: PMN has been described with the nearest-neighbor formalism developed for metallic alloys [11], and the Nb/vacancy distribution in $Nb_{1-x}CoSb$ has been described with a pair-interaction Hamiltonian combined with structural relaxation around vacancies [9]. These simple modeling methods, however, fail to capture the full complexity of the kinetically trapped disorder often found in these materials. To overcome this limitation, we develop a new method for fitting diffuse scattering from such crystals using the combination of techniques described below.

To test the method, we use the binary Fe–20 at.% Al alloy as a model system. Despite its simple average structure, its disorder is relatively complicated, featuring non-trivial substitutional correlations and strong relaxation around the substituted atoms — the same structural features that make relaxors and thermoelectrics difficult to model. Fe-Al has recently been chosen by the single-crystal diffuse scattering consortium (Oak Ridge National Laboratory, ETH Zürich, the University of Bremen and the University of Erlangen) as the first reference material for the comparison of diffuse scattering modeling techniques, as it is also convenient experimentally: Fe and Al give high contrast in both X-ray and neutron scattering, absorb weakly, and do not activate appreciably under neutron irradiation. This work is the first analysis in the series. The other groups will model the same data using Monte Carlo, reverse Monte Carlo, and mean-field approaches.

Modeling the size effect in such a system requires atoms to be displaced from their ideal lattice sites, which in turn requires the energy of an arbitrary atomic configuration to be evaluated. Density functional theory (DFT) does this accurately, but is far too expensive for the tens of thousands of atoms needed to reproduce a diffuse scattering pattern. We therefore use universal machine-learned interatomic potentials (uMLIPs), which have emerged as a low-cost solution for relaxing large-scale structural ensembles with DFT-like results [12]. These potentials are pre-trained on large quantum-mechanical databases, and models such as those from the Universal Models for Atoms (UMA) family perform competitively with system-specific potentials [13].

The resulting workflow combines Yell refinement, histogram reweighting, and Monte Carlo simulations with UMA relaxation to create a 16,000-atom model of Fe-Al. This work is the first to show that such an approach reproduces the experimental diffuse scattering across multiple reciprocal-space planes and is directly transferable to more complex materials with substitutional and size-effect correlations.

2 Theory

2.1 Diffuse Scattering. The total scattering intensity (I_{total}) from a crystal can be mathematically decoupled into a Bragg component (I_{Bragg}) and a diffuse scattering component ($I_{diffuse}$). [1]

$$I_{total} = I_{Bragg} + I_{diffuse} \tag{1}$$

The Bragg term encompasses information about the average crystal structure, while the diffuse term holds information about structural perturbations from the average. Borie and Sparks [4] formalized that the diffuse scattering term can be further expanded into components representing short-range order (I_{SRO}), size effects (I_{SE}), and thermal diffuse scattering (I_{TDS}).

$$I_{diffuse} = I_{SRO} + I_{SE} + I_{TDS} \tag{2}$$

2.2 Short-Range Order. Short-range order describes how atoms order with regard to their nearest neighbors. The local chemical composition can be simulated with an Ising-like Hamiltonian [14],

$$E_{\text{configuration}} = J_1 \sum_{\langle ij \rangle_1} s_i s_j + J_2 \sum_{\langle ij \rangle_2} s_i s_j + \dots \tag{3}$$

where s_i and s_j represent spin site-occupancy operators (+1 for Fe, −1 for Al), $\langle ij \rangle_n$ the spin-pair summation over the nth nearest neighbors, and J_n the interaction parameter at that nth shell. The sign of J determines whether the system will favor unlike-pair bonds (SRO) or like-pair bonds (clustering/phase separation) [15]. For the Fe-Al alloy, we adopt the convention that positive Js favor SRO over clustering.

2.3 Size-Effect. Fe and Al atoms have different metallic radii. As such, elastic strain is created when these atoms are forced to sit on an ideal, rigid lattice. When allowed to move off the grid during relaxation, the optimizer shifts atoms to relieve this strain and consequently creates static atomic displacements, or the "size-effect".

2.4 Thermal Diffuse Scattering. To get a proper comparison against the experimental data, which was measured at room temperature, thermal energy must be added to the system. While not entirely representative, an isotropic B-factor approximates thermal motion by modeling the vibrations of individual atoms while ignoring correlations between them.

2.5 Correlations in 3D-ΔPDF Space. It is much easier to analyze correlations from scattering in real space than in reciprocal space. The Yell software accomplishes this by taking the Fourier transform of the diffuse scattering signal, which is calculated by subtracting the Bragg scattering (average structure) from the total scattering. From the shape of this new signal—the three-dimensional difference pair distribution function (3D-ΔPDF)—we can interpret substitutional, size effect, and thermal displacement correlations [17, 18].

Substitutional correlations appear as symmetric positive or negative peaks on the 3D-ΔPDF map: positive peaks indicate a favoring of like-pair occupancies, while negative peaks indicate the opposite. This is due to the difference in probability of finding like-pairs compared to the random crystal structure with correlation—now, if we know the identity of a certain atom, the probability of its neighbors changes depending on whether like-pairs or unlike pairs are favored. Thus, the resulting signal in the pair distribution function (PDF) is greater or less than the average (Patterson function) depending on whether the pair occurs more often than random.

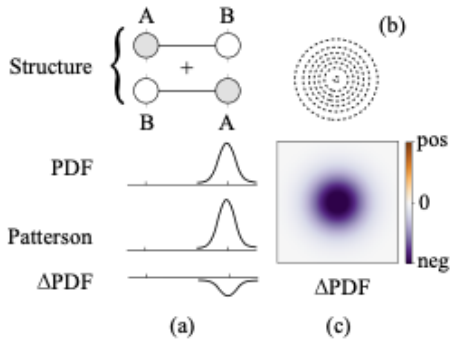


Figure 1a. Fingerprint of a negative substitutional correlation. Reproduced with permission from [18]

The size effect appears as either positive-negative or negative-positive features in the ΔPDF space. The interatomic distances between pairs of atoms differ based on the pair's composition. This shows up in the PDF as two partly overlapping peaks with different heights, where the taller one corresponds to the pair with the stronger scattering (heavier) atom. When crystallographically averaged across the structure, the Patterson ends up as a single broad peak centered at an intermediate distance, so after subtraction, we get an antisymmetric signal. And, since we know the positive signal sits at the separation distance of the heavier atom, we can discern the direction of the local relaxation.

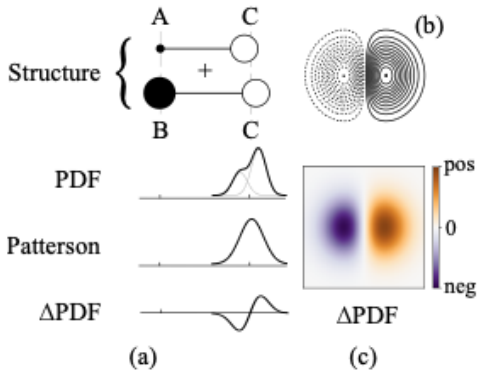


Figure 1b. Fingerprint of a size effect correlation. Reproduced with permission from [18]

Thermal displacement correlations appear as a negative-positive-negative or positive-negative-positive signal, depending on whether atoms tend to displace in the same or opposite direction. In the first case, the separation between atoms fluctuates less than if they moved independently. Thus, the width of the PDF is narrower, and when the average is subtracted, the resulting signal has an alternating negative-positive-negative pattern. If instead the atoms moved in opposite directions, the reverse occurs, and we get the inverted positive-negative-positive signal.

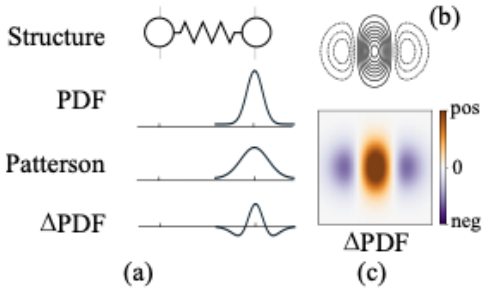


Figure 1c. Fingerprint of a negative-positive-negative thermal displacement correlation. Reproduced with permission from [18]

2.6 Histogram Reweighting. In order to refine the interaction parameter J against experimental data, it is useful to calculate derivatives such that we can relate changes in J to changes in structural configuration. There is a linear relationship between configurational energy and J (Equation 3) ,

which we can rewrite as,

$$E_{\mathbf{J}} = \sum_k J_k S_k, \tag{4}$$

where S_k is the spin-pair sum of shell k and the conjugate to J_k. This allows us to formulate a linear response expression for structure susceptibility χ_{kl} [19]

$$\chi_{kl} = \frac{\partial \langle dp_l \rangle}{\partial J_k}. \tag{5}$$

where dp_l is the deviation from the like-pair joint probability (0.8*0.8 = 0.64) in shell l, or the substitutional correlation at shell l. This stems from thermodynamics, where we can take a derivative of free energy to calculate values of conjugate variables. Taking the derivative with respect to J_k shows how SRO changes in shell l when changing the Js of shell k.

To derive this relationship, we first start by representing the expectation value of an observable A (in this case, dp_l) at a J estimated from configurations at a reference J_0

$$\langle A \rangle = \frac{\sum_i A_i w_i}{\sum_i w_i}, \qquad w_i = \exp\Big[-\tfrac{1}{T}\big(E_{\mathbf{J}}(i) - E_{\mathbf{J}_0}(i)\big)\Big]. \tag{6}$$

Differentiating the expectation with respect to J_k, we have

$$\frac{\partial \langle A \rangle}{\partial J_k} = -\frac{1}{T}\big(\langle A S_k \rangle - \langle A \rangle \langle S_k \rangle\big) = -\frac{1}{T}\,\mathrm{Cov}(A, S_k). \tag{7}$$

Because the interaction parameter J was defined with spin variables ($s = \pm 1$) while dp was calculated with occupation variables ($\sigma = 0/1$) to match Yell's parameterization of Fe–Fe pair correlations, $s = 2\sigma - 1$ and pair interaction terms $\sigma_i \sigma_j$ scale by a factor of 4:

$$s_i s_j = 4\sigma_i \sigma_j - 2\sigma_i - 2\sigma_j + 1. \tag{8}$$

The middle terms are also constant since the alloy concentration never changes, so our spin pair sum reduces to $S_k = 4n_k + C$, where n_k is the count of Fe-Fe pairs in shell k. Our susceptibility expression then becomes

$$\chi_{kl} = -\frac{4}{T}\,\mathrm{Cov}(dp_l, n_k). \tag{9}$$

To find the ΔJ which minimizes the residual norm between the simulated pair correlations and the experimentally fitted ones (dp_{diff}), we must solve the equation

$$\boldsymbol{\chi}\,\Delta\mathbf{J} = \mathbf{dp}_{\mathrm{diff}}, \tag{10}$$

where χ is the susceptibility matrix. Since it's overdetermined in our use case (we solve for 12 Js with 25 shells), we multiply both sides by χ^{T} to get

$$\boldsymbol{\chi}^{\mathsf{T}}\boldsymbol{\chi}\,\Delta\mathbf{J} = \boldsymbol{\chi}^{\mathsf{T}}\mathbf{dp}_{\mathrm{diff}}. \tag{11}$$

However, χ is only the first derivative, so the linear approximation neglects curvature and degrades as ΔJ grows, so steps easily overshoot. Thus, to damp the step size, we apply a Marquardt-scaled regularization term [20] with lambda fixed at 1.0

$$\Big(\boldsymbol{\chi}^{\mathsf{T}}\boldsymbol{\chi} + \lambda\,\mathrm{diag}(\boldsymbol{\chi}^{\mathsf{T}}\boldsymbol{\chi})\Big)\Delta\mathbf{J} = \boldsymbol{\chi}^{\mathsf{T}}\mathbf{dp}_{\mathrm{diff}}, \tag{12}$$

Solving this final equation for ΔJ leaves us with the values by which to increment J.

3 Methods

3.1 Experimental data. Single-crystal diffraction data were collected at the ESRF ID28 beamline (experiment HC-5160) using monochromatic X-rays of wavelength 0.6976 Å. The sample was polished on four sides in the shape of a square prism. Data were recorded at room temperature by rotating the crystal 360° about the ϕ axis in 0.1° steps, with a total of 3600 frames per run. To broaden the reciprocal-space coverage of the Pilatus 1M detector, two datasets were taken with the detector at 19° and 48°. Raw frames were indexed with XDS [21] and reconstructed into a three-dimensional reciprocal-space volume with Meerkat [22]. The two volumes were then merged and symmetry-averaged over the m-3m Laue group to improve the statistics of the diffuse scattering signal.

3.2 Computational Environment. All data processing and simulations were conducted on an NVIDIA RTX A6000 GPU. We primarily used a Python-based environment (3.12.3) with pandas (3.0.3), numpy (2.4.6), and matplotlib (3.10.9) for data handling, alongside specialized scientific libraries Atomic Simulation Environment (ASE) (3.28.0) [23], FAIR Chemistry (2.20.0), gemmi (0.7.5) [24], gemmi_disorder (0.1.0), and h5py (3.16.0). Non-Python software included Yell (1.3.79) and Meerkat (0.39).

3.3 Yell refinement. We perform a 3D-ΔPDF analysis on the experimental data. First, we subtract the Bragg peaks from our experimental h5 file using the Meerkat package, with a punch and fill inner punch radius of 9.5 and outer filling shell of 10.0. The experimental diffuse intensity was then mapped onto a $200 \times 200 \times 200$ reciprocal-space grid spanning ± 5 r.l.u with a 0.05 r.l.u. step, and symmetrized under the m-3m Laue group. Refinements employed Yell's approximate scattering calculation on a $64 \times 64 \times 64$ FFT grid with analytical derivatives, via the Ceres least-squares minimizer. The simulation extended as far out as 220 shells to (10,10,-4). All adp and se values were set to 0; only the scale factor and the self-correlation adp of 0.0055 were set to ensure a self-variance baseline.

3.4 Refining Js from Pair Correlations. To create a physical model from the calculated Yell correlations, we first ran a least-squares regression between the shell pair counts and energies of a set (300) of structures to determine the starting set of Js. Due to computational constraints, these structures were $10 \times 10 \times 10$ supercells (instead of $20 \times 20 \times 20$) subject to a Monte Carlo (T = 4.2) simulation and UMA relaxation. However, these preliminary configurations did not reproduce the experimental Js well. Several factors may have contributed to this, such as inaccuracies in the initial interaction parameters, limitations of the UMA model, or differences between the experimental disorder and the equilibrium configurations sampled by the Monte Carlo simulation. In particular, if the experimental short-range order is partially kinetically frozen during cooling, an equilibrium model may not fully capture the observed correlations. The present results do not distinguish between these possible explanations, and these preliminary J parameters served solely as starting values for the subsequent histogram reweighting optimization. All interaction parameters and simulation temperatures are expressed in a consistent reduced-unit convention, with $k_B = 1$.

Then, a rigid-lattice Monte Carlo (MC) simulation based on the Metropolis algorithm [25] (T = 0.2 reduced units, chosen empirically based on preliminary trial runs) was executed with 14 million steps on a set of $20 \times 20 \times 20$ body-centered cubic (BCC) supercells using the preliminary Js and

containing 16,000 atoms with a randomized concentration of 80% Fe and 20% Al. The residual norm

$$R_{norm} = \sqrt{\sum_{shell} (dp_{diff})^2} \tag{13}$$

and per-shell histogram for the first 25 shells were computed to assess the fit. We chose to use 12 Js based on the drop-off in substitutional correlation magnitude between the 12th and 13th shell.

A damped update step ΔJ was calculated from the susceptibility (section 2.6), and a new set of 300 MC simulations was again run with the updated parameters. This process was repeated until the difference in norm between steps was less than .0001, which took $\sim$8 iterations.

3.5 Relaxation with UMA. The final 300 structures generated from histogram reweighting were relaxed with the uma-s-1p2 checkpoint and OMat task head from UMA. This was paired with the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) optimizer, with a convergence criterion that the maximum residual force on each atom fell below 0.01 eV/Å. The UMA model was selected due to its computational efficiency at large scales [13].

3.6 Calculating Diffuse Scattering. To eliminate statistical noise and accurately capture ensemble behavior, all sets of structures were processed using the `average_diffuse` function from the gemmi_disorder package [16]. Unrelaxed structures used the parameters blur=0.04, crop=2, and b_iso=0.0 to evaluate pure static disorder, while for the relaxed structures, an isotropic B-factor of 0.24 Å^2 was introduced via the b_iso parameter to model thermal motion. Truncation artifacts also arose when b_iso was kept at 0.0. The two other parameters, blur and crop, help account for the FFT discretization noise and any edge effects that may arise from the finite reciprocal-space grid, respectively. All values were chosen empirically based on alignment with experimental profiles. The resulting maps were symmetrized under the $m\bar{3}m$ Laue group using Meerkat, after which the *(hk0)* plane and five distinct *(xkl)* planes were sliced out for visual and quantitative analysis.

4 Results

4.1 Experimental Diffuse Scattering. Figure 2 shows experimental diffuse scattering in the reciprocal space *(hk0)* plane, with arrows indicating typical correlation features. Substitutional correlations appear as symmetric blobs centered between the principal Bragg peaks (arrow a), size effect correlations show up as intensity asymmetries along certain reciprocal space directions (e.g., the $[1\bar{1}0]$ direction) (arrow b) and around Bragg peaks, and thermal displacement correlations appear as rodlike structures (arrow c) centered at the Bragg peaks.

4.2 3D-ΔPDF. We first attempted to refine the substitutional (dp), size-effect (se), and thermal displacement (adp) correlations simultaneously in Yell for the first six shells. However, the refinement was ill-conditioned and produced an unphysical model; the calculated scattering displayed negative intensities near the Bragg peaks. One possible explanation is the low resolution of the dataset, which makes it difficult to distinguish the similar features of substitutional and thermal displacement contributions in reciprocal space. To reduce this ambiguity, we increased the masking region around the Bragg peaks and refined just the substitutional correlations, as they dictate the underlying structure. Yell-refined values for the first 25 shells are listed in Appendix Table A1, and the *(uv0)* plane of the Yell-refined and experimental 3D-ΔPDF is shown in Figure 3.

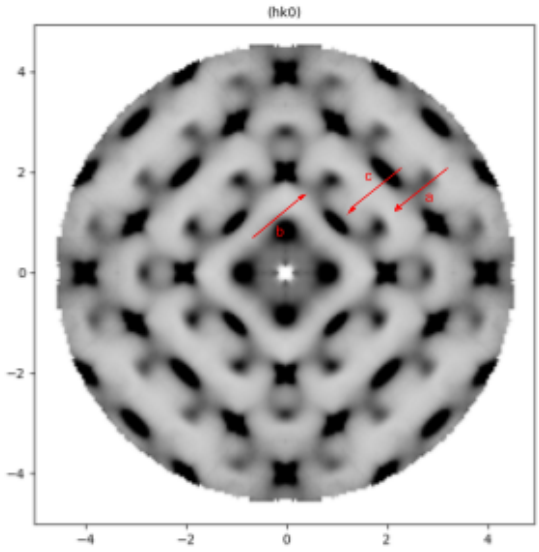


Figure 2. Experimental diffuse scattering in the (hk0) plane. Arrows point to substitutional (a), size effect (b), and thermal displacement (c) correlation features

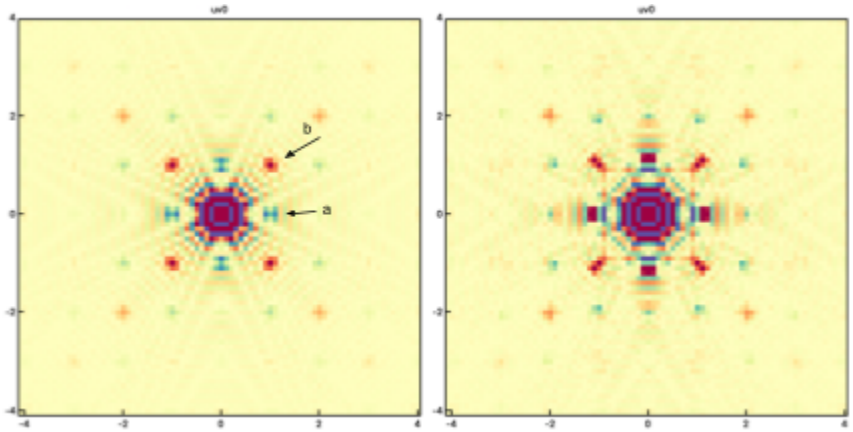


Figure 3. Yell-refined (left) and experimental (right) 3D-ΔPDF maps of the *(uv0)* plane. Arrow a indicates a negative substitutional correlation and arrow b indicates a positive substitutional correlation

The peaks line up with the BCC lattice sites (which are also the shell vectors). In the *(uv0)* plane, we clearly see signals at vector multiples of [100] and [110]. For example, we have a negative peak at 1, 0, 0 (arrow a) and a positive peak at 1, 1, 0 (arrow b). Additionally, correlations lose intensity farther out from the center; this decay is a sign of SRO.

The refinement converged to R_w=26%, suggesting that substitutional correlations are a large portion of the diffuse signal. This agreement is supported visually—the Yell-refinement of pure substitutional correlations captures much of the experimental 3D-ΔPDF map.

4.3 Comparison to Literature Warren-Cowley (WC) Parameters. The first five WC parameters [26], calculated by taking $\frac{dp}{c(1-c)}$, where c is the fraction of Fe, are listed in Table 1 along with those reported by Pierron-Bohnes et al. [27] for an alloy of 19.5 at.% Al.

The 1st and 3rd shells, which are the two largest correlations, both agree within 15% of literature

Table 1. Comparison of first five Warren-Cowley parameters for Fe–20 at.% Al (this work) and Fe–19.5 at.% Al [27]

	Yell-refined	Pierron-Bohnes
α_1	−0.073	−0.065(7)
α_2	−0.052	0.008(8)
α_3	0.071	0.080(6)
α_4	0.0098	none listed
α_5	−0.011	0.005(3)

WC values. The 2nd and 5th differ in sign, though the 5th value is an order of magnitude less than the leading correlations. As for the 2nd, we note that α_2 is the least well-constrained parameter in the study, being consistent with zero within its stated error. Additionally, differences in sample composition and thermal history, which determine the temperature at which SRO is frozen, may also contribute to this discrepancy.

4.4 Results from Histogram Reweighting. The initial least-squares regression (Fig. 4) from the $10 \times 10 \times 10$ supercells has an $R^2 = 0.946$, indicating that the regression effectively captured the relation between model-predicted and UMA-predicted energy.

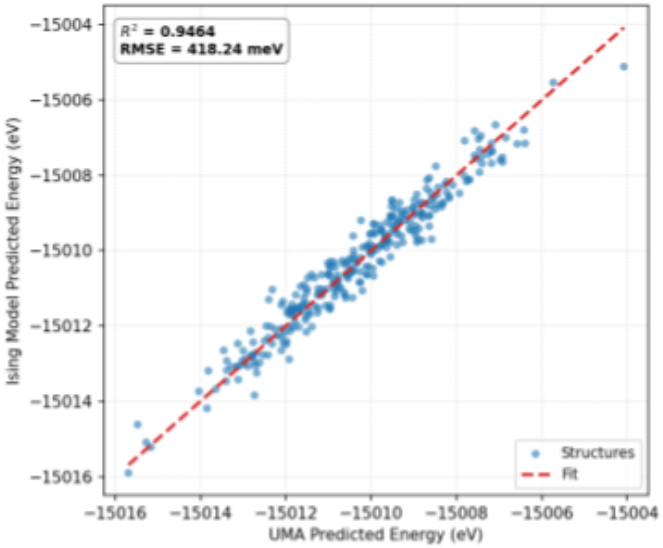


Figure 4. Linear regression ($R^2 = 0.946$, RMSE = 418 meV) between Ising-model-calculated and UMA-calculated energies of 300 structures ($10 \times 10 \times 10$ supercell of 2,000 atoms).

The initial and final histograms from the reweighting are shown in Figure 5, and their corresponding J values are listed in Appendix Table A2. Notably, the residual norm was reduced from 0.012 to 0.0034, confirming that reweighting was able to push the simulation's dp values closer to experimental values.

4.5 Simulated Diffuse Scattering Maps. The diffuse scattering map of the *(hk0)* plane (Fig. 6, left) after pure Monte-Carlo without any relaxation displays periodic, symmetric diffuse features centered between the principal Bragg peaks at h+k = odd, confirming the successful encoding of SRO into

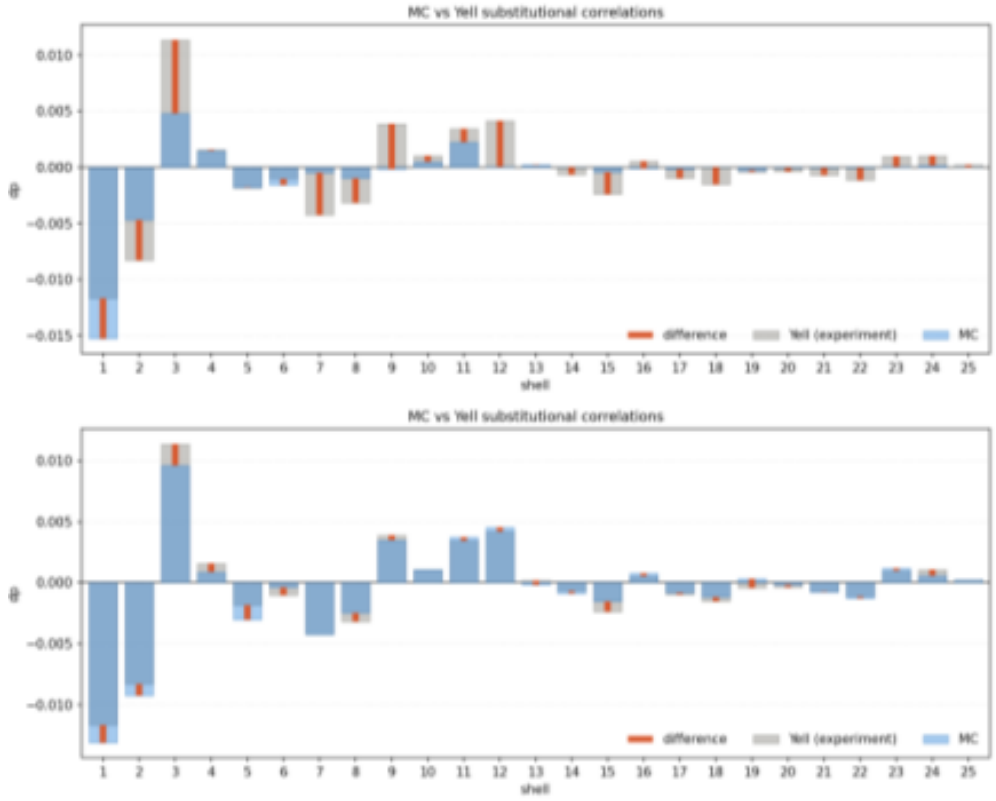


Figure 5. MC and YeH substitutional correlations at each shell before (top) and after (bottom) reweighting. R_{norm} was reduced from 0.012 to 0.0034.

the structure. After relaxation with UMA, the size effect becomes apparent due to the symmetry breaking, which shows up as an intensity asymmetry around Bragg peaks and specific reciprocal space directions (Fig. 6, middle). This indicates that the uMLIP is able to simulate physically meaningful static displacements. Although already quite close to experimental data (Fig. 6, right), the simulation is still missing the rod-like streaks radiating from Bragg peaks typical of TDS. Because our approximation is based on an isotropic atomic displacement model, these correlated vibrational features are naturally absent from the simulation.

Furthermore, the simulation captures the changing symmetry patterns across *(xkl)* planes (Fig. 7), confirming its ability to model the structure in all dimensions of 3D reciprocal space, not just a single favorable cross-section. Simulated and experimental planes differ slightly in their l values, e.g., 0.40 vs 0.38, due to the step sizes of the experimental HDF5 grid.

4.6 Simulation Limitations. While the implementation of this pipeline was successful, its scope remains bounded by limitations inherited from the UMA model. Because UMA is trained on Density Functional Theory (DFT) data, phenomena that go beyond DFT, such as finite-temperature magnetic fluctuations or explicit electron-phonon coupling, cannot be explicitly captured. Furthermore, machine-learned potentials' reliability degrades beyond their training domain, so they can introduce non-physical artifacts when extrapolating [28]; thus, rigorous validation and careful discernment are still essential. Another limitation is that correlated thermal motion is not explicitly modelled, entering only through the b_iso parameter, so the workflow as presented would require extension before being applied to a material in which thermal contributions dominate.

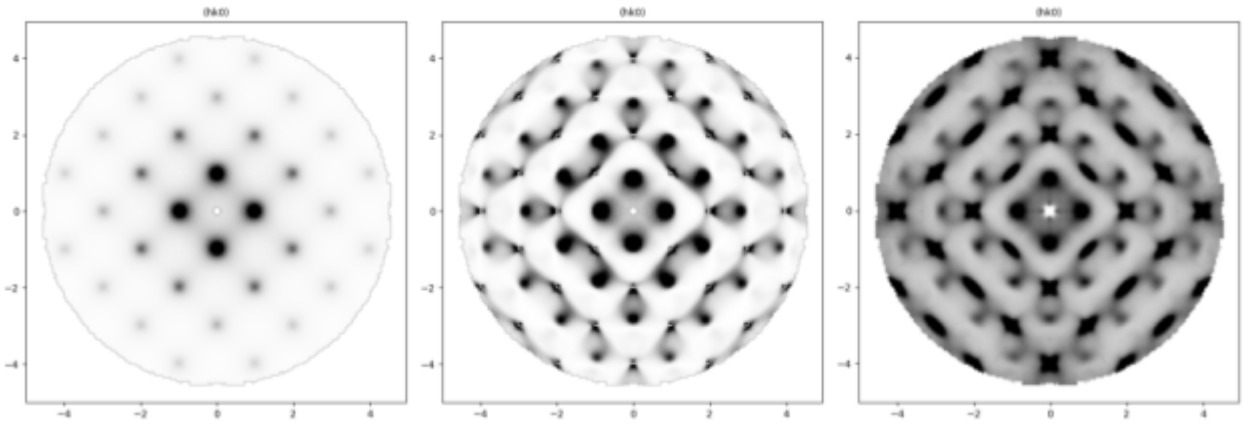


Figure 6. Simulated diffuse scattering maps in the *(hk0)* plane; from left to right, pure MC diffuse scattering, diffuse scattering after relaxation, and experimental data

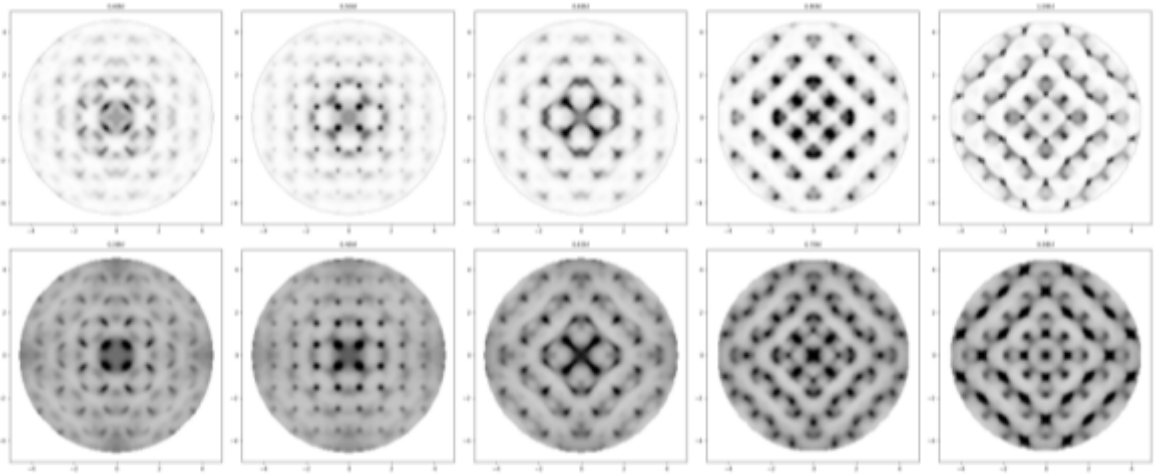


Figure 7. Simulated diffuse scattering maps in five distinct *(xkl)* planes; from left to right, $x = 0.3833$, 0.5, 0.6167, 0.8, 1

The structural relaxations were performed using a lattice parameter of (a=2.867 Å), corresponding to pure Fe, whereas the indexed lattice parameter of the experimental crystal is (2.898 Å). The relaxation cell is therefore approximately 1.1% smaller than the indexed cell. This compression may affect the magnitude and spatial distribution of the static displacements, and hence the calculated size-effect contribution to diffuse scattering. The resulting displacement patterns should therefore be interpreted with this lattice-parameter mismatch in mind.

5 Conclusion

This work establishes a workflow combining 3D-ΔPDF analysis, histogram reweighting, Monte Carlo simulation, and uMLIPs to model local disorder in single-crystal alloys. Rather than refining all correlation types simultaneously—which proved ill-conditioned, as substitutional and displacement correlations produce nearly indistinguishable diffuse scattering with low-resolution data—the workflow divides the problem: substitutional correlations are refined from experiment,

while the displacements are generated physically through relaxation with a uMLIP. Applied to a binary Fe–20 at.% Al system, this approach reproduced chemical SRO and the size effect across full 3D reciprocal space. Notably, the size-effect asymmetries emerged directly from relaxation, without any system-specific fitting, suggesting that uMLIPs can accurately reproduce the static atomic displacements that diffuse scattering is highly sensitive to. These methods are directly transferable to more complex materials whose diffuse scattering carries comparable substitutional and displacive contributions.

The GitHub repository for this work can be located at `https://github.com/cynthia-wen/disordered-materials-fe-al-alloy`.

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Appendix

Table A1. Yell-refined dps for the first 25 shells, to three significant figures

Shell	*dp*	Shell	*dp*
1	−0.0116	14	−0.000674
2	−0.00828	15	−0.00237
3	0.0114	16	0.000537
4	0.00158	17	−0.000989
5	−0.00183	18	−0.00154
6	−0.00103	19	−0.000407
7	−0.00423	20	−0.000398
8	−0.00317	21	−0.000721
9	0.00389	22	−0.00114
10	0.00106	23	0.000985
11	0.00346	24	0.00107
12	0.00416	25	0.000209
13	0.000163		

Table A2. J values before and after reweighting for the first 12 shells, to three significant figures

Shell	Initial Js	Final Js (after reweighting)
1	0.0420	0.0340
2	0.0197	0.0211
3	−0.00204	−0.00487
4	−0.00141	−0.000246
5	0.00378	0.00242
6	0.00221	0.00331
7	0.000222	0.00558
8	0.000486	0.00165
9	−0.000097	−0.00237
10	−0.000390	−0.00237
11	−0.00321	−0.00546
12	0.000659	−0.00150

Table A3. Notation and definitions used throughout the report.

Symbol	Definition
s_i	Spin variable at site i: $+1$ for Fe and -1 for Al.
σ_i	Occupancy variable: 1 for Fe and 0 for Al, with $s_i = 2\sigma_i - 1$.
J_n	Pair-interaction coefficient for the nth neighbor shell in the Ising-like Hamiltonian.
S_n	Sum of spin-pair products over pairs in shell n.
n_n	Number of Fe–Fe pairs in shell n, using the pair-counting convention defined in the implementation.
dp_n	Shell-resolved substitutional correlation: $P(\mathrm{Fe}-\mathrm{Fe}) - c_{\mathrm{Fe}}^2$.
χ_{kl}	Susceptibility matrix element describing the response of correlation dp_l to a change in interaction parameter J_k.
ΔJ_k	Update to the interaction parameter for shell k during histogram reweighting.
T	Reduced simulation temperature, expressed in the same reduced energy convention as the interaction parameters; used in the Boltzmann factor $(\exp(-E/T))with(k_B = 1)$.
λ	Dimensionless damping coefficient in the regularized least-squares update.
R_{norm}	Unweighted Euclidean norm of the residuals between simulated and target shell correlations.
R_w	Weighted residual factor reported by the Yell refinement; see the refinement software's definition.
α_n	Warren–Cowley short-range-order parameter for shell n.
`adp`	Atomic-displacement parameter used in the Yell refinement.
`se`	Size-effect correlation parameter used in the Yell refinement.
`blur`	Smoothing parameter used in the diffuse-scattering calculation.
`crop`	Parameter controlling treatment of finite-grid edge effects.
`b_iso`	Isotropic displacement parameter used in the diffuse-scattering calculation, in $\mathrm{Å}^2$ if interpreted as an isotropic crystallographic B factor.